

Daily Grid Forensics: SPP

24 constraint(s) | window 2026-08-07 → 2026-08-13

1. Executive Summary

Date	Constraint	DA (\$)	RT (\$)	Driver	
2026-08-07	Infrandl-spence1	0	-9500	NPPD Zonal Load	WAU WAUE Wind
2026-08-11	Indobson-gano	0	-8220	SECI Zonal Load	Wind generation in OKGE and SPS
2026-08-10	Innebrcty-sub3456	-2707	-3925	JASPER7 JASPER_MORGAN A LN (110958) outage	OPPD Zonal Load
2026-08-07	Int_brmghm-lbrtyst5	0	-6357	St. Joseph - Fairport 345 kV Line Outage Coal / KCPL	KCPL Zonal Load
2026-08-13	Inweav-tallgras	-5941	0	East Eureka - Vaughn 115 kV line outage Wind / OKG Zone	WR Zonal Load
2026-08-11	Incraig-lenexa	-4656	-897	Wind/WR WR	KCPL Zonal Load Wind/OKG OKGE
2026-08-11	Inholcom1-grdnct1	-189	-4246	SECI Zonal Load	SPP System Load
2026-08-13	xfmrcimarron-cimarron	-3555	-610	Wind generation in OKGE	OKGE Zonal Load
2026-08-10	xfmrredwlo1-redwlo1	-1040	-2963	NPPD Zonal Load	System-wide Wind Generation
2026-08-10	Instilwell-redel5	-2916	-914	MPS Zonal Load	Jasper - Morgan 345kV Line Outage
2026-08-10	Inipcjeff4-lieberm	-344	-3186	CSWS Zonal Load	Wind/OKG OKGE Wind/SPS SPS
2026-08-12	Intallgras-butl	-3399	0	Coal in KCP	Wind in MEC Wind in WR
2026-08-07	Inmdwst-frnkl1	-256	-2774	WFEC zonal load Pittsburg - Valliant 345kV Line outage	OKGE zonal load

2026-08-08	Ingord-maiz	-491	-1925	WR Zonal Load (Peak Load conditions) Butler to Otter Creek 138kV Transmission Line (eq_num 113340) MacArthur - Seneca Line 69kV (eq_num 31491)
2026-08-13	Inmoun-jade	-1429	-587	OKG OKGE Wind SPS SPS Natural Gas SPS SPS Wind
2026-08-13	xfmrnowst1-nowst1_rev23	-1989	0	OKGE Zonal Load OKGE Wind
2026-08-10	Inbuln-kel	-1745	-181	System Load
2026-08-12	Inksc10-farb	-83	-1792	Wind in WR zone (Slate Creek Wind) Wind in OKGE zone Flat Ridge 2 (VIOLA7 345)
2026-08-11	Indavejohn-lar_rivr	0	-1835	SPP Extreme Summer Peak Load Wind Generation (OKGE, WR, MEC)
2026-08-13	Infulton_c-pat_wx	0	-1723	CSWS Generation (Fulton, Turk) Redispatch System Wind Under-forecast (Redispatch to CSWS Gen)
2026-08-11	Inharbine-fairbury	0	-1693	NPPD Zonal Load
2026-08-13	Inpecan1-agncy1	-1630	0	Wind/OKG OKGE EES EES Zonal Load SOU SOUC Zonal Load
2026-08-09	Infrankln5-litc	-1522	0	WR Zonal Wind EES Zonal Load Misc Generation
2026-08-13	xfmrturk_pp-turk_pp	-110	-1302	CSWS Zonal Load EES EES Natural Gas OKG OKGE Wind

Infrandl-spence1

DA: \$0.00 | RT: \$-9500.00

2. Constraint Topology

- **Monitored Element:** Infrandl-spence1
- **Market Context:** SPP
- **Profile:** The Infrandl-spence1 constraint refers to the 115 kV transmission line connecting the Fort Randall Substation in Charles Mix County, South Dakota, to the Spencer Substation in Boyd County, Nebraska. This line is part of the Nebraska Public Power District (NPPD) system within the Southwest Power Pool (SPP) market.

3. Explanation Section (Analysis Date: 2026-08-07)

3.1 Summary: The Fort Randall - Spencer 115 kV constraint bound heavily in RT (-9500) but not in DA (0) on August 7, 2026. This divergence was driven by a significant under-forecast of NPPD zonal load, which is the dominant loading zone for this constraint. Actual NPPD load came in an average of 561 MW higher than the DA forecast, with a maximum absolute error of 1021 MW. This higher-than-expected load on the low-gen side aggravated the constraint in real-time, leading to the severe binding. No transmission outages were present to contribute to the congestion.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 4.7x
- **Analyst Note:** On 2026-08-07, NPPD zonal load was in Peak Load status (avg 3,410.3 MW), wind generation was Low Wind (avg 14,137.9 MW), and solar generation was Very Low Solar (avg 1,237.9 MW).

3.5 DA/RT Reconciliation: Channel 1 explains the gap: NPPD zonal load (the dominant loading zone) came in significantly above the day-ahead forecast (average error +561 MW, max absolute error +1021 MW at HE 24). This under-forecast in DA meant the DA model did not anticipate the load-driven congestion that materialized in RT, pricing RT (-9500) well

above DA (0). Channel 2 checked: OutageAgent returned no scheduled or realtime outages for the target date, ruling out a topology divergence.

3.6 External Research Context:

Historical SPP Market Monitoring reports identify this constraint as a recurring regional flowgate located along the South Dakota–Nebraska border. No publicly indexed web sources confirm a specific transmission outage or severe weather event on August 7, 2026 tied to this binding constraint.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-08 to 2026-08-07)

5. Reference Links

- [Identifies the constraint as the Fort Randall - Spencer 115 kV line and lists the contingency.](#)
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► 7. Dayzer Charts (click to expand)

Indobson-gano

DA: \$0.00 | RT: \$-8220.00

2. Constraint Topology

- **Monitored Element:** Indobson-gano
- **Market Context:** SPP
- **Profile:** The constraint is on the 115 kV transmission line from Gano Substation to Dobson Substation. Both substations are located in the vicinity of Garden City, Finney County, Kansas, operating within the SPP market.

3. Explanation Section (Analysis Date: 2026-08-11)

3.1 Summary: The Gano-Dobson 115 kV constraint in SPP bound in real-time on August 11, 2026, driven by an extreme peak in SECI zonal load. The SECI load hit an annual record of 1,262.9 MW during severe summer heat, overloading the line for 11 hours. The day-ahead model failed to price the constraint because it under-forecasted SECI load by up to 89 MW during the peak hours. No transmission outages contributed to the event.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 5.0x
- **Analyst Note:** On 2026-08-11, SECI zonal load reached its annual peak of 1,262.9 MW (100th percentile), driving 11 real-time binding hours (hr 10-20) with a maximum constraint shadow price severity of \$1,295.46/MWh under moderate wind conditions.

3.5 DA/RT Reconciliation: The DA/RT divergence (DA 0, RT -8220) is explained by Channel 1: local load forecast error. At hour 16, actual SECI zonal load was 1260 MW against a day-ahead forecast of 1173 MW (+87 MW miss). The peak error was +89 MW at hour 17. This significant under-forecast of load on the aggravating side caused the constraint to bind in RT while remaining unpriced in DA. Channel 2 was checked via OutageAgent, which returned no transmission outage suspects, ruling out an outage schedule discrepancy.

3.6 External Research Context:

External research confirms the Dobson-Gano 115 kV line is a known historical constraint in the SPP footprint (Mid-Kansas Electric Company / Sunflower Electric Power Association area), frequently identified in SPP High Priority Incremental Load Studies and DISIS reports. No external reports specifically cited the August 11, 2026 binding event, consistent with it being a short-term load-driven operational limit rather than a major grid failure.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural

5. Reference Links

- [Historical context on the Dobson-Gano constraint from SPP planning studies.](#)
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► 7. Dayzer Charts (click to expand)

Innebrcty-sub3456

DA: \$-2707.00 | RT: \$-3925.00

2. Constraint Topology

- **Monitored Element:** Innebrcty-sub3456
- **Market Context:** SPP
- **Profile:** The constraint 'Innebrcty-sub3456' represents the 345 kV transmission line connecting the Nebraska City Power Station (identified as Substation 3458) to the Sarpy County Station (identified as Substation 3456) within the Omaha Public Power District (OPPD) zone. This line is a critical path for delivering baseload generation from Nebraska City north towards the Omaha load center.

3. Explanation Section (Analysis Date: 2026-08-10)

3.1 Summary: The Nebraska City - Sarpy County 345kV constraint (Innebrcty-sub3456) in the OPPD zone bound heavily on August 10, 2026, with RT shadow prices reaching -\$3925/MW. The constraint is structurally predisposed to bind during peak load conditions, and OPPD load was at peak levels (avg 2639 MW). The acute trigger for the severe binding was the forced outage of the Jasper-Morgan 69kV line, which started at 08:00 on the event day and increased the risk of binding by 10x. The DA/RT divergence (RT priced higher than DA) is explained by both the unscheduled RT outage of the Jasper-Morgan line and OPPD load coming in above the day-ahead forecast.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
JASPER7 JASPER_MORGAN A LN	2026-08-10 08:00:00	-	1.2%	1.4%	11.47x	240.7 mi
SUB3459 TR3 TR3 XF	2026-08-10 07:30:00	2026-08-11 15:00:00	1.6%	0.6%	2.45x	6.6 mi
COOPER FAIRPT-COOPR A LN	2026-04-10 08:30:00	2026-10-16 16:00:00	5.1%	8.2%	1.29x	19.3 mi
MRYVL_SJ MRYVLBRADD16_1 1 LN	2026-08-10 12:24:00	2026-08-14 12:24:00	12.6%	2.3%	1.03x	39.3 mi
SPLT_RT SPLT_WHITE34_1 1 LN	2026-07-29 07:00:00	2026-08-20 16:00:00	1.5%	1.4%	1.00x	170.9 mi

3.4 Weather Drivers:

- **Condition:** Peak Load

- **Impact Multiplier:** 3.2x
- **Analyst Note:** On 2026-08-10, OPPD zonal load was at Peak Load (2638.58 MW avg), wind generation was Med Wind (17936.62 MW avg), and solar generation was Very Low Solar (1255.94 MW avg).

3.5 DA/RT Reconciliation: RT priced higher than DA (-3925 vs -2707). Channel 1 explains part of the gap: actual OPPD load exceeded the DA forecast by an average of 125 MW (peak error +102 MW), which aggravates the constraint in RT. Channel 2 also explains the gap: the Jasper-Morgan 69kV line outage (eq_num 110958) was present in the realtime records but absent from the scheduled records, meaning DA did not price its congestion impact.

3.6 External Research Context:

No external sources confirmed the specific outage or constraint activation on August 10, 2026. General reports note the Nebraska City-Sub 3456 345kV line as a historically binding constraint in SPP during peak conditions.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Short-Term (2025-08-11 to 2026-08-10)

5. Reference Links

- [Historical context on constraint binding](#)
- [Historical context on constraint binding](#)

► 7. Dayzer Charts (click to expand)

Int_brmghm-lbrtyst5

DA: \$0.00 | RT: \$-6357.00

2. Constraint Topology

- **Monitored Element:** Int_brmghm-lbrtyst5
- **Market Context:** SPP
- **Profile:** This constraint monitors the 161 kV transmission line between the Liberty South Substation (located near 2000 Birmingham Rd in Liberty, MO) and the Birmingham Substation (located near 7th & Milwaukee in Birmingham, MO) within the KCPL (Evergy) zone in Missouri. The constraint is likely activated for the contingent loss of the Missouri City to Birmingham line.

3. Explanation Section (Analysis Date: 2026-08-07)

3.1 Summary: The Liberty South - Birmingham 161 kV line bound in the SPP real-time market due to a combination of a chronic transmission outage and an acute load forecast miss. The long-standing outage of the St. Joseph - Fairport 345 kV line elevated base flows on the underlying 161 kV network, acting as a structural predisposition. On the target date, actual KCPL zonal load came in below the day-ahead forecast. Because KCPL load structurally relieves this constraint, the lower-than-expected load removed expected relief, driving flows higher and causing a massive real-time price spike (-\$6,357) with no day-ahead congestion.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
ST_JOE FAIRPT-STJOE A LN	2026-06-08 08:00:00	2026-08-21 16:00:00	6.3%	874.1%	5.58x	42.0 mi
COOPER FAIRPT-COOPR A LN	2026-04-10 08:30:00	2026-10-16 16:00:00	1.2%	283.8%	3.06x	55.0 mi

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 4.8x
- **Analyst Note:** On 2026-08-07, KCPL zonal load was at Peak Load (2,534.8 MW avg), wind generation was Low Wind (14,137.9 MW avg), and solar generation was Very Low Solar (1,237.9 MW avg).

3.5 DA/RT Reconciliation: Channel 1 explains the divergence. KCPL zonal load is a structural relieving driver for this constraint (averaging -77.4 MW of relief). On the target date, actual KCPL load came in below the day-ahead forecast (daily average miss of -84 MW, with a maximum absolute error of -288 MW at HE 11). Lower load in a relieving zone removes expected relief, which increases flow and prices RT above DA.

3.6 External Research Context:

External research confirms the Birmingham - Liberty City 161 kV line is a long-standing SPP flowgate identified in historical transmission planning studies. No specific real-time outage or trip was reported publicly for the target date, aligning with the internal finding that the primary outage was a long-standing scheduled event and the acute trigger was a load forecast miss.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-08 to 2026-08-07)

5. Reference Links

- [Historical aggregate study referencing the Birmingham - Liberty City 161 kV constraint.](#)
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► 7. Dayzer Charts (click to expand)

Inweav-tallgras

DA: **\$-5941.00** | RT: **\$0.00**

2. Constraint Topology

- **Monitored Element:** Inweav-tallgras
- **Market Context:** SPP
- **Profile:** The Weaver - Tallgrass constraint is a 138 kV transmission line located in Butler County, Kansas, owned by Evergy Kansas Central (EKC). It connects the Weaver Substation (near US 400 and SW Indianola Rd in Andover) to the Tallgrass Substation (near I-35 and SW Buffalo Rd south of El Dorado). This line has been identified for rebuild projects to mitigate regional congestion.

3. Explanation Section (Analysis Date: 2026-08-13)

3.1 Summary: The Weaver - Tallgrass 138 kV line in the Evergy Kansas Central (WR) zone experienced a massive day-ahead to real-time divergence on August 13, 2026, binding at -\$5,941/MW in the DA market but clearing at \$0 in real-time. The primary driver of the day-ahead congestion was the scheduled outage of the East Eureka - Vaughn 115 kV line, which has a 21.4% LODF on the monitored element. This outage was included in the day-ahead model but was cancelled or deferred in real-time, completely removing the congestion in the RT market. High WR zonal load (peaking at 6,439 MW) provided the base flow, but the acute trigger for the DA binding was the phantom outage.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
EEUR 4225 1 LN	2026-08-08 13:34:00	2026-08-21 16:00:00	21.4%	61.2%	2.28x	34.1 mi

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 3.4x
- **Analyst Note:** On 2026-08-13, WR zonal load was at Peak Load (5,450.2 MW avg), wind generation was at Peak Wind (19,386.3 MW avg), and solar generation was at Very Low Solar (1,138.3 MW avg).

3.5 DA/RT Reconciliation: The DA shadow price was -\$5,941/MW while RT was \$0. This divergence is explained by Channel 2 (Realtime line outages differed from the day-ahead

schedule). The East Eureka - Vaughn 115 kV line outage (eq_num 31118) was present in the scheduled outage records but absent from the realtime outage records, meaning the day-ahead market priced congestion around an outage that did not occur in real-time. Channel 1 was checked and rejected: the WR zonal load forecast error was 0 MW at the peak hour (16:00).

3.6 External Research Context:

SPP planning documents identify the Weaver-Tallgrass 138 kV line as a recognized path prone to regional flowgate congestion. No specific operational notice or trip report was found for August 13, 2026, aligning with the finding that the outage was a day-ahead phantom event.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-14 to 2026-08-13)

5. Reference Links

- [Identifies the Weaver - Tallgrass line as a recognized path prone to regional flowgate congestion in SPP planning documents.](#)
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► 7. Dayzer Charts (click to expand)

Incraig-lenexa

DA: \$-4656.00 | RT: \$-897.00

2. Constraint Topology

- **Monitored Element:** Incraig-lenexa
- **Market Context:** SPP
- **Profile:** The Craig - Lenexa 161 kV constraint refers to the 161 kV transmission line connecting the Craig Substation (located at 10859 Woodland Rd, Olathe, KS) and the Lenexa Substation (located at 15730 W 95th St, Lenexa, KS). This line is a key component of the Evergy (formerly KCPL) transmission network in Johnson County, Kansas, often monitored by SPP for congestion relief.

3. Explanation Section (Analysis Date: 2026-08-11)

3.1 Summary: The Craig-Lenexa 161 kV constraint bound on August 11, 2026, driven primarily by high wind generation in the WR and OKGE zones, coupled with peak load in the KCPL zone. The day-ahead market priced the constraint significantly higher than real-time due to a large wind under-forecast, which resulted in less loading-side wind generation in real-time than scheduled.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Wind
- **Impact Multiplier:** 2.3x
- **Analyst Note:** On 2026-08-11, conditions in SPP/NPPD featured Peak Wind generation (18,107.49 MW avg), Peak Zonal Load (3,555.21 MW avg), and Very Low Solar generation (1,227.03 MW avg).

3.5 DA/RT Reconciliation: DA priced the constraint significantly higher than RT (DA -4656 vs RT -897). Channel 1 explains the gap: system-wide wind (a proxy for the material WR/OKGE wind load drivers) came in 1,789 MW BELOW the day-ahead forecast at HE 16 (13,725 MW actual vs 15,513 MW forecast). This shortfall in loading-side wind relieved the constraint in real-time relative to the DA model. KCPL load came in 323 MW above forecast,

which would have priced RT higher, but the massive wind shortfall dominated. Channel 2 checked: no outage records returned for the target date. Channel 3 checked: no internal rating change detected.

3.6 External Research Context:

External sources discuss the Craig-Lenexa 161 kV line in the context of regional transmission planning and rebuild projects (e.g., Evergy/NextEra filings). No specific real-time outage or event was reported for August 11, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural

5. Reference Links

- [Mentions the Craig-Lenexa 161 kV line rebuild in regional planning documents.](#)
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► 7. Dayzer Charts (click to expand)

Inholcom1-grdnct1

DA: \$-189.00 | RT: \$-4246.00

2. Constraint Topology

- **Monitored Element:** Inholcom1-grdnct1
- **Market Context:** SPP
- **Profile:** The HOLCOM1 10140 A LN constraint involves a 115 kV transmission line connecting the Holcomb Substation and the Garden City Substation in Finney County, Kansas, within the Southwest Power Pool (SPP) market.

3. Explanation Section (Analysis Date: 2026-08-11)

3.1 Summary: The Holcomb - Garden City 115 kV line (Inholcom1-grdnct1) bound in SPP real-time on August 11, 2026, driven by extreme peak load conditions in western Kansas. The SECI zonal load reached its annual maximum peak of 1,262.9 MW during an SPP summer heat wave, pushing flows on the line to its thermal limit under the loss of the Holcomb - Setab 345 kV line. The massive RT price spike (-4246 vs DA -189) was caused by SECI load coming in 89 MW above the day-ahead forecast at the peak hour, forcing expensive real-time redispatch.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 5.0x
- **Analyst Note:** On 2026-08-11, SECI zonal load reached its annual maximum peak of 1,262.9 MW (97.1st daily percentile) and SPP total load reached 61,325 MW (95.5th daily percentile), while wind generation was moderate (71.3rd percentile) and solar was elevated (82.0th percentile).

3.5 DA/RT Reconciliation: RT price (-4246) was significantly higher than DA price (-189). Channel 1 explains this: SECI zonal load came in 89 MW ABOVE the day-ahead forecast at peak hour 17 (actual 1263 MW vs forecast 1173 MW). Since SECI is a loading zone, this

under-forecast of load in DA led to higher actual flows in RT, driving the RT price spike.
Channel 2 checked: no transmission outages found.

3.6 External Research Context:

WebSearchAgent found no specific outage or event on the Holcomb - Garden City line on August 11, 2026. However, SPP operating logs indicate that the West Balancing Authority Area (which includes western Kansas) was under heightened operational advisories during this period, including a Conservative Operations Advisory through August 10 and a Resource Advisory starting August 13 due to high peak load forecasts and low wind generation.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-12 to 2026-08-11)

5. Reference Links

- [Confirms heightened operational advisories in the SPP West BAA due to high peak load and low wind generation.](#)
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► 7. Dayzer Charts (click to expand)

xfmrcimarron-cimarron

DA: \$-3555.00 | RT: \$-610.00

2. Constraint Topology

- **Monitored Element:** xfmrcimarron-cimarron
- **Market Context:** SPP
- **Profile:** The Cimarron - Cimarron constraint refers to a transformer (likely 345/138 kV) within the Cimarron Substation, located near Yukon in Canadian County, Oklahoma. This substation is a key node in the Oklahoma Gas & Electric (OKGE) transmission system, connecting to the Mathewson Substation (near Piedmont) and the Draper Substation (near Oklahoma City) via 345 kV lines. It facilitates the flow of wind energy from western Oklahoma to load centers in the east.

3. Explanation Section (Analysis Date: 2026-08-13)

3.1 Summary: The Cimarron 345/138 kV transformer constraint bound severely in the Day-Ahead market (m-value -3555) compared to Real-Time (-610) on August 13, 2026. This constraint is a structural bottleneck for wind generation moving from western Oklahoma to eastern load centers. The primary driver was high wind generation in the OKGE zone combined with peak load conditions. The significant DA/RT divergence was caused by an over-forecast of wind generation in the Day-Ahead market (averaging 3,330 MW higher than actuals), which led the DA model to expect much higher flows and price the constraint accordingly. Transmission outages were evaluated but ruled out as false positives due to negligible flow impacts.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
SUNLN 8889 A LN	2026-08-13 06:00:00	2026-08-13 10:30:00	3.6%	0.1%	20.22x	17.9 mi
SAGE 8888 A LN	2026-08-13 06:00:00	2026-08-13 10:30:00	3.6%	0.1%	20.22x	16.4 mi
WDWRDEHV XF2 XF2_H XF	2026-02-03 12:29:00	2026-12-31 17:00:00	1.2%	0.3%	1.17x	106.5 mi
CLINT_CY 2198 A LN	2026-03-06 08:00:00	2026-08-31 16:00:00	3.5%	0.0%	1.35x	54.5 mi

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 4.1x

- **Analyst Note:** On 2026-08-13, OKGE zonal load averaged 6,465 MW (Peak Load), wind generation averaged 19,386 MW (Peak Wind), and solar generation averaged 1,138 MW (Very Low Solar).

3.5 DA/RT Reconciliation: The DA shadow price (-3555) was significantly more negative than RT (-610). This is explained by Channel 1: Wind generation was over-forecasted in DA. The day-ahead wind forecast averaged 22,841 MW while actual wind averaged 19,511 MW (a miss of -3,330 MW, peaking at -5,841 MW in HE 18). Since wind is a strong aggravating driver for this constraint, the over-forecast in DA led to a much higher expected flow and a more severe shadow price in the DA market. Channel 2 (outages) was checked and ruled out as no significant outages were found.

3.6 External Research Context:

SPP Market Analysis indicates the Cimarron 345/138 kV transformer is a key binding constraint driven by strong wind generation in the southern plains combined with high load hours. No specific acute events were reported for the target date.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-14 to 2026-08-13)

5. Reference Links

- [Confirms the structural nature of the constraint driven by wind and load.](#)
- [Joint stakeholder briefing materials noting the constraint's historical congestion patterns.](#)

► 7. Dayzer Charts (click to expand)

xfmrredwlo1-redwlo1

DA: \$-1040.00 | RT: \$-2963.00

2. Constraint Topology

- **Monitored Element:** xfmrredwlo1-redwlo1
- **Market Context:** SPP
- **Profile:** The constraint REDWLO1 T1 T1_H XF monitors the Red Willow 345/115 kV transformer (T1) located at the Red Willow Substation in Nebraska. Although the equipment is listed as a 'Line from REDWILO3 to REDWLO1', the bus designations (345 kV and 115 kV) and the 'XF' in the constraint name indicate this is a step-down transformer within the NPPD zone of the SPP market.

3. Explanation Section (Analysis Date: 2026-08-10)

3.1 Summary: The xfmrredwlo1-redwlo1 constraint (Red Willow 345/115 kV transformer) bound severely in the SPP real-time market on August 10, 2026, driven by extreme peak load in the NPPD zone. The DA/RT divergence (RT shadow price -2963 vs DA -1040) was caused by a significant under-forecast of NPPD zonal load day-ahead, with actual load exceeding the forecast by up to 461 MW. This was compounded by lower-than-forecasted system-wide wind generation, which reduced relieving flows. External research confirms SPP was under a Conservative Operations Advisory due to high peak loads and variable energy resource uncertainty.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 4.8x
- **Analyst Note:** On 2026-08-10, NPPD zonal load was at Peak Load (3,468.80 MW average), while solar generation was Very Low Solar (1,255.94 MW average) and wind generation was at Med Wind (17,936.62 MW average).

3.5 DA/RT Reconciliation: RT shadow price (-2963) was much more severe than DA (-1040). Channel 1 explains this: NPPD zonal load was significantly under-forecasted day-

ahead. At peak hour (HE 16), actual load was 3978 MW vs DA forecast of 3642 MW (+336 MW error). Max absolute error was +461 MW at HE 20. Since NPPD load aggravates the constraint, the higher RT load drove the RT shadow price higher. System-wide wind was also under-forecasted (actual average 17,937 MW vs forecast 20,689 MW, max error -3,910 MW at HE 21), reducing relieving wind generation. Channel 2 checked: no relevant realtime-only outages found.

3.6 External Research Context:

SPP was operating under elevated grid alerts across its Western Balancing Authority Area (West BAA) during this period. SPP terminated an Energy Emergency Alert 1 (EEA1) for the West BAA at 12:01 a.m. CT on August 10, 2026, but remained under a Conservative Operations Advisory through August 11, 2026 due to forecasted high peak loads, elevated generation outages, and variable energy resource (wind/solar) uncertainty.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural

5. Reference Links

- [Confirms SPP was under elevated grid alerts \(Conservative Operations Advisory\) due to high peak loads and variable energy resource uncertainty on the target date.](#)
- [Confirms SPP West stayed in Conservative Operations through Tuesday due to high loads.](#)

► 7. Dayzer Charts (click to expand)

Instilwell-redel5

DA: \$-2916.00 | RT: \$-914.00

2. Constraint Topology

- **Monitored Element:** Instilwell-redel5
- **Market Context:** SPP
- **Profile:** The Stilwell - Redel constraint refers to the 161 kV transmission line connecting the Stilwell Substation (located at 6300 W 191st St, Stilwell, KS) and the Redel Substation (located at 4409 W 159th St, Overland Park, KS). This line is part of the Evergy (formerly KCP&L) transmission system in Johnson County, Kansas, within the SPP market.

3. Explanation Section (Analysis Date: 2026-08-10)

3.1 Summary: The Stilwell - Redel 161 kV line bound in the SPP market on August 10, 2026, driven primarily by peak load conditions in the MPS zone and high wind generation in the OKG zone. The constraint is structurally predisposed to bind under these conditions, as confirmed by historical sensitivity analysis and SPP planning studies. The acute trigger was the combination of peak load and the forced outage of the Jasper - Morgan 345kV line, which aggravated the flow. The DA model priced the constraint significantly higher than RT due to an over-forecast of both MPS load and system wind.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
JASPER7 JASPER_MORGAN A LN	2026-08-10 08:00:00 -		1.0%	21.3%	11.83x	102.7 mi
SHEF 422 1 LN	2026-07-06 07:00:00	2026-08-12 17:00:00	1.2%	0.1%	3.60x	86.7 mi
SEDPLT 11070 A LN	2026-07-13 10:00:00	2026-08-12 16:00:00	2.0%	0.5%	1.53x	77.3 mi

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 2.0x
- **Analyst Note:** On 2026-08-10, MPS zonal load was in the Peak Load bucket (avg 1521.67 MW), wind generation was Med Wind (avg 17936.62 MW), and solar generation was Very Low Solar (avg 1255.94 MW).

3.5 DA/RT Reconciliation: DA shadow price (-2916) was significantly higher in magnitude than RT (-914). Channel 1 explains this gap: MPS zonal load was over-forecast in DA (actual

averaged 47 MW below forecast, with a max absolute error of -162 MW at HE 12). Additionally, system wind (a major loading factor) was over-forecast by an average of 2753 MW. Both over-forecasts led the DA model to price the constraint higher than what materialized in RT. Channel 2 checked: Jasper-Morgan 345kV outage was present in RT but not scheduled, which would have priced RT above DA, the wrong direction to explain the gap.

3.6 External Research Context:

External research confirms the Stilwell - Redel 161 kV line is a recognized transmission bottleneck in the Evergy/SPP system, featured in SPP planning and interconnection studies calling for rebuilds to increase capacity. No specific external news confirmed an acute event on August 10, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-11 to 2026-08-10)

5. Reference Links

- [Confirms the Stilwell - Redel 161 kV line is a recognized transmission bottleneck in the SPP system.](#)
-

► 7. Dayzer Charts (click to expand)

Inipcjeff4-lieberm

DA: \$-344.00 | RT: \$-3186.00

2. Constraint Topology

- **Monitored Element:** Inipcjeff4-lieberm
- **Market Context:** SPP
- **Profile:** The Lieberman - IPC Jefferson 138 kV line is a transmission line located in the AEP (SWEPCO) zone, connecting the Lieberman Power Plant substation in Mooringsport, Louisiana, to the IPC Jefferson substation in Jefferson, Texas.

3. Explanation Section (Analysis Date: 2026-08-10)

3.1 Summary: The Lieberman - IPC Jefferson 138 kV constraint bound in SPP on August 10, 2026, driven primarily by extreme summer peak load in the CSWS zone. Historical analysis confirms this constraint is strictly driven by high load, binding exclusively during top-quintile peak load conditions. On the target date, SPP system load peaked above 60,000 MW (94.7th percentile) and CSWS zonal load reached the 91st percentile, pushing flow on the line to its 212 MW rating. Wind generation in OKGE and SPS provided secondary loading contributions. The significant DA/RT divergence (RT -3186 vs DA -344) remains unexplained by available forecast-vs-actual data or outages, suggesting a possible unverified realtime switching solution or localized realtime deviation.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 5.0x
- **Analyst Note:** On 2026-08-10, SPP total load averaged 50,052 MW (94.7th percentile, peaking at 60,280 MW in HE17) and CSWS zonal load averaged 7,765 MW (91.0th percentile), with total wind at 17,937 MW (70.5th percentile) and total solar at 1,256 MW (82.5th percentile).

3.5 DA/RT Reconciliation: RT priced significantly higher than DA (-3186 vs -344). Channel 1 checked: CSWS zonal load at HE18 was 20 MW below forecast (actual 9780 MW vs forecast

9800 MW), and system-wide wind was 3517 MW below forecast. If wind is an aggravating driver, lower wind would price DA above RT, which is the wrong direction. Channel 2 checked: OutageAgent returned no scheduled or realtime outages. Channel 3 checked: No capacity change detected in Panorama. Channels 4 and 5 (switching/virtuals) remain unverified. The divergence is unexplained by available data.

3.6 External Research Context:

A search of SPP public reporting and industry news yielded no specific outage or operational report explicitly tied to the Inipcjeff4-lieberm constraint binding on August 10, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-11 to 2026-08-10)

5. Reference Links

- [SPP RTBM Binding Constraints](#)
-

► 7. Dayzer Charts (click to expand)

Intallgras-butl

DA: **\$-3399.00** | RT: **\$0.00**

2. Constraint Topology

- **Monitored Element:** Intallgras-butl
- **Market Context:** SPP
- **Profile:** The constraint monitors the 138 kV transmission line connecting the Tallgrass Substation to the Butler Substation in El Dorado, Butler County, Kansas, within the Southwest Power Pool (SPP) market.

3. Explanation Section (Analysis Date: 2026-08-12)

3.1 Summary: The Tallgrass - Butler 138 kV line bound in the SPP Day-Ahead market on August 12, 2026, driven by peak load conditions that necessitated high dispatch of local coal and gas generation. Historical analysis confirms the constraint is overwhelmingly driven by HIGH_LOAD conditions. Long-term structural decomposition identifies Coal in KCP (+45.6 MW avg) and Wind in MEC/WR as the primary loading drivers. During the target date's peak load (WR zone reached 96th percentile), the increased output from these generators to serve load pushed the post-contingency flow on the line to its limit. The constraint did not bind in Real-Time due to a massive under-forecast of wind and an over-forecast of net-relieving WR load, both of which reduced RT congestion.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 4.6x
- **Analyst Note:** On 2026-08-12, WR zonal load (96th percentile, averaging 5,610 MW and peaking at 6,835 MW) and SPP total load were both in Peak Load conditions, driving Day-Ahead congestion on Intallgras-butl across 12 hours with a peak severity of \$497.77/MWh.

3.5 DA/RT Reconciliation: DA priced at -\$3399/MWh while RT did not bind (\$0). Channel 1 explains this massive divergence through two forecast misses that both reduced RT

congestion: (1) System-wide wind was under-forecast by 3,398 MW at the binding hour (HE 17). Since wind in MEC, WR, and OKGE is a primary loading driver, lower wind in RT provided less loading than DA expected. (2) WR zonal load actual exceeded forecast by 388 MW at HE 17. Since WR load is a net relieving zone (-43 MW avg impact), higher load in RT provided more relief than DA expected. Both factors priced DA above RT. Channel 2 checked: no outage drivers found. Channel 3 checked: no rating change detected.

3.6 External Research Context:

External research found no specific outage or acute grid event on August 12, 2026. SPP planning documents (such as the 20-Year Assessment and Notifications to Construct) reference the Tallgrass 138 kV substation and long-term congestion management along the Butler-Tallgrass corridor, confirming it as a known structural bottleneck, but these do not point to a discrete event on the target date.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-13 to 2026-08-12)

5. Reference Links

- [SPP 20-Year Assessment Report](#) referencing the Tallgrass 138 kV substation and long-term congestion trends.
 - [SPP Notification to Construct](#) for the Tallgrass 138 kV project.
-

► 7. Dayzer Charts (click to expand)

Inmdwst-frnkln1

DA: \$-256.00 | RT: \$-2774.00

2. Constraint Topology

- **Monitored Element:** Inmdwst-frnkln1
- **Market Context:** SPP
- **Profile:** The constraint is the 138 kV transmission line connecting the Midwest Substation in Midwest City, Oklahoma, to the Franklin Substation in Norman, Oklahoma. The Franklin Substation is located at the southwest corner of 48th Avenue NW and Franklin Road in Norman. The Midwest Substation serves the Midwest City area, likely located near the city center or Tinker Air Force Base.

3. Explanation Section (Analysis Date: 2026-08-07)

3.1 Summary: The Midwest - Franklin 138 kV line bound in SPP due to high load in the WFEC zone (aggravating) combined with a massive loss of relief from the OKGE zone. The DA/RT divergence (-256 DA vs -2774 RT) was driven by actual OKGE load coming in 1,639 MW below the day-ahead forecast at the binding hour. Because OKGE load relieves the constraint, this shortfall removed expected relief in realtime, driving flow higher and pricing RT significantly above DA. An aggravating outage on the Pittsburg - Valliant 345kV line was present but ended early in RT, which would have relieved the constraint and cannot explain the RT price spike.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
PITTSB9 2160 A LN	2026-08-04 07:30:00	2026-08-07 17:00:00	1.8%	4.2%	1.61x	98.2 mi

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 4.4x
- **Analyst Note:** On 2026-08-07, OKGE zonal load was in the Peak Load bucket (avg 5,435.61 MW), wind generation was Low Wind (avg 14,137.92 MW), and solar generation was Very Low Solar (avg 1,237.90 MW).

3.5 DA/RT Reconciliation: RT priced significantly above DA (-2774 vs -256). Channel 1 explains the gap: actual load in the relieving OKGE zone came in 1,639 MW below the day-

ahead forecast at the binding hour (HE 17). This massive shortfall in a relieving zone removed expected relief in realtime, driving flow higher and pricing RT above DA. Channel 2 checked and rejected: the aggravating Pittsburg - Valliant 345kV outage ended early in RT (08:18 vs scheduled 17:00), which would have relieved RT and priced DA above RT, the wrong direction. Channel 3 unverified (no internal rating move data).

3.6 External Research Context:

Searches did not turn up any public reporting, regulatory filings, or news records explaining a specific real-world event, outage, or driver for the Inmdwst-frnkln1 constraint on August 7, 2026. No external confirmation found.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-08 to 2026-08-07)

5. Reference Links

- [General constraint dataset](#)
 - [SPP documents and filings](#)
 - [SPP curtailments](#)
-

► 7. Dayzer Charts (click to expand)

Ingord-maiz

DA: \$-491.00 | RT: \$-1925.00

2. Constraint Topology

- **Monitored Element:** Ingord-maiz
- **Market Context:** SPP
- **Profile:** Constraint likely representing the transmission path or flowgate between the Longford Substation (Clay County, KS) and the Maize Substation (Sedgwick County, KS) within the Evergy (Western Resources) zone of the SPP market. Longford is a major 345kV hub, while Maize is a key 138kV substation near Wichita.

3. Explanation Section (Analysis Date: 2026-08-08)

3.1 Summary: The Ingord-maiz (Longford - Maize 138kV) constraint bound in the SPP market on August 8, 2026, primarily driven by high electric demand in the WR zone reaching Peak Load conditions. This structural load driver was exacerbated by the aggravating outage of the Butler to Otter Creek 138kV line. The significant RT > DA price divergence is partially explained by WR zonal load exceeding the day-ahead forecast at the peak hour, and the cancellation of a scheduled relieving outage (MacArthur - Seneca 69kV) in real-time, which removed expected relief from the RT market.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
BUTL BUTL_OTTERCRK A LN	2026-08-07 04:28:00	2026-08-12 04:00:00	10.9%	0.0%	1.68x	33.2 mi
MACA MACAOATV69_1 1 LN	2026-02-23 11:23:00	-	3.6%	0.2%	0.52x	10.3 mi
MACA MACASENE69_1 1 LN	2026-05-04 10:00:00	2026-09-18 15:00:00	6.0%	0.2%	0.29x	11.2 mi

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 1.9x
- **Analyst Note:** On 2026-08-08, WR zonal load reached Peak Load conditions (avg 4636.06 MW), accompanied by Low Wind (avg 12023.18 MW) and Very Low Solar (avg 1260.55 MW).

3.5 DA/RT Reconciliation: RT shadow price (-1925) was significantly higher than DA (-491). Channel 1 checked: WR zonal load actual exceeded the day-ahead forecast by 140 MW at peak hour 18, which prices RT above DA. Channel 2 checked: The scheduled outage of the

relieving MacArthur - Seneca 69kV line (eq_num 31491) was not present in the realtime list (cancelled/not taken), which prices RT above DA as RT lacked the expected relief. However, the aggravating Butler to Otter Creek 138kV line (eq_num 113340) was also scheduled but not in realtime, which would price DA above RT. Channel 3 checked: no internal rating change data available.

3.6 External Research Context:

No specific real-time outage or grid event report for August 8, 2026 was confirmed for the SPP constraint Ingordmaiz in public SPP filings or grid logs. SPP issued broader regional operational advisories (e.g., Resource Advisory for the West BAA), but these are not confirmed evidence of a specific event on this constraint on the target date.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-09 to 2026-08-08)

5. Reference Links

- [General SPP grid conditions and congestion reporting; no specific event confirmed for the target date.](#)
 - [SPP congestion by constraint portal.](#)
-

► 7. Dayzer Charts (click to expand)

Inmoun-jade

DA: \$-1429.00 | RT: \$-587.00

2. Constraint Topology

- **Monitored Element:** Inmoun-jade
- **Market Context:** SPP
- **Profile:** The Jade - Moundridge 115 kV line is a transmission line located in Kansas within the SPP market. It connects the Jade Substation in Hillsboro, KS to the Moundridge Substation in Moundridge, KS.

3. Explanation Section (Analysis Date: 2026-08-13)

3.1 Summary: The Jade - Moundridge 115 kV line (Inmoun-jade) in SPP bound heavily in the Day-Ahead market on August 13, 2026, driven by near-record wind generation. Structurally, the constraint is loaded by wind generation in the OKG and SPS zones. On the target date, SPP system-wide wind reached 24,179 MW during light overnight load hours, saturating the export corridor. The Day-Ahead market over-forecasted wind by 1,380 MW at hour 5, leading to a much higher DA shadow price (-1429) compared to Real-Time (-587).

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Wind
- **Impact Multiplier:** 1.5x
- **Analyst Note:** SPP total wind averaged 19,421 MW (peaking at 24,179 MW in HE1, 98th percentile of annual distribution), total load averaged 50,832 MW (ranging from 41,717 to 60,094 MW), and solar generation averaged 1,091 MW.

3.5 DA/RT Reconciliation: Channel 1 (wind forecast error) explains the gap. DA shadow price was -1429, RT was -587. Wind is the primary aggravating driver for this constraint. On 2026-08-13, SPP system-wide wind (used as a proxy for local wind) was significantly over-forecast in the Day-Ahead market. At hour 5, DA forecast was 25,270 MW while RT actual was 23,890 MW (a -1,380 MW miss); the daily average miss was -3,420 MW. Because wind

loads the constraint, the higher DA wind assumption priced DA congestion much higher than RT. Channel 2 (outages) was checked but no outage records were returned.

3.6 External Research Context:

No specific outage, transmission trip, or real-world grid event for the SPP Inmoun-jade constraint on August 13, 2026 was identified in public transmission planning reports, RTO Insider, or GridStatus data. No external confirmation found.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-15 to 2026-08-13)

5. Reference Links

- No verified source links found.
-

► 7. Dayzer Charts (click to expand)

xfmrnowst1-nowst1_rev23

DA: \$-1989.00 | RT: \$0.00

2. Constraint Topology

- **Monitored Element:** xfmrnowst1-nowst1_rev23
- **Market Context:** SPP
- **Profile:** Profiling failed.

3. Explanation Section (Analysis Date: 2026-08-13)

3.1 Summary: The xfmrnowst1-nowst1_rev23 constraint (NOWST1 345/138kV transformer) bound in the Day-Ahead market on August 13, 2026, driven by extreme summer peak load in the OKGE zone and a massive over-forecast of wind generation. OKGE load is the dominant structural driver of this constraint, pulling power through the transformer. On the target date, SPP system load reached the 99.7th percentile. The DA/RT divergence (DA bound at -1989, RT cleared at 0) was caused by a 5,626 MW over-forecast of wind at the binding hour (HE 17). Since wind in OKGE is a loading driver, the DA model placed significantly more flow on the constraint than materialized in RT, causing it to bind only in DA.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 5.0x
- **Analyst Note:** SPP system load reached extreme summer peak levels (averaging 59,702 MW during hours 16-19, peaking at 60,094 MW in the 99.7th percentile), accompanied by moderate wind generation (~15,512 MW) and elevated solar generation (~2,208 MW).

3.5 DA/RT Reconciliation: DA priced at -1989, RT at 0 (DA > RT). Channel 1 explains the gap: System-wide wind (a proxy for local OKGE wind, which is a loading driver) was over-forecast by 5,626 MW at HE 17 (actual 15,207 MW vs forecast 20,832 MW). This over-forecast placed more flow on the constraint in the DA model, causing it to bind, while the lower actual wind in RT relieved it. Channel 2 checked: OutageAgent found no scheduled or

realtime outages on the target date. Channel 3 checked: No capacity change detected in Panorama.

3.6 External Research Context:

No public operational log, ISO report, or news coverage explicitly confirms the SPP constraint xfmrnwst1-nowst1_rev23 bound on the target date of August 13, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-14 to 2026-08-13)

5. Reference Links

- [SPP market report context](#)
 - [SPP stakeholder briefing](#)
-

► 7. Dayzer Charts (click to expand)

Inbuln-kel

DA: \$-1745.00 | RT: \$-181.00

2. Constraint Topology

- **Monitored Element:** Inbuln-kel
- **Market Context:** SPP
- **Profile:** The Inbuln-kel constraint refers to the 161 kV transmission line connecting the Bullion Substation and the Kelso Substation in Missouri. The Bullion Substation is a new switching station built to split the existing Kelso - Miner 161 kV line, located near Benton in Scott County. The Kelso Substation is located in Kelso Township, Missouri. This line is part of the Ameren/Sikeston (SIKE) zone within the SPP market.

3. Explanation Section (Analysis Date: 2026-08-10)

3.1 Summary: The Bullion - Kelso 161 kV line bound in the SPP market on 2026-08-10 primarily due to peak load conditions. Historically, this constraint is sensitive to both high solar and high load. On the target date, load reached peak levels, driving the day-ahead shadow price to -1745. However, actual system load came in below the day-ahead forecast, which relieved the constraint in real-time and resulted in a much lower RT shadow price of -181. Transmission outages were evaluated but ruled out due to insignificant flow impacts.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 4.6x
- **Analyst Note:** On 2026-08-10, WR zonal load was at Peak Load (avg 5289.17 MW), wind generation was at Med Wind (avg 17936.62 MW), and solar generation was at Very Low Solar (avg 1255.94 MW).

3.5 DA/RT Reconciliation: Channel 1 explains the divergence: System-wide load came in below the day-ahead forecast (average error -431 MW, max absolute error -1033 MW at hour 11). Since load aggravates this constraint, the lower actual load in real-time relieved the congestion, resulting in a much lower RT shadow price (-181) compared to DA (-1745).

Channel 2 checked: the INDEPCO5 outage was present in both scheduled and realtime, so no schedule mismatch occurred.

3.6 External Research Context:

No external confirmation found. A targeted search of SPP market reports and constraint filings yielded no verifiable records or published outage events for LNBULN-KEL on August 10, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Short-Term (2025-08-10 to 2026-08-10)

5. Reference Links

- [SPP Real-Time Balancing Market data.](#)
 - [SPP Integrated Transmission Planning.](#)
-

► 7. Dayzer Charts (click to expand)

Inksc10-farb

DA: \$-83.00 | RT: \$-1792.00

2. Constraint Topology

- **Monitored Element:** Inksc10-farb
- **Market Context:** SPP
- **Profile:** The Farber - Sumner County No. 10 constraint is a 138 kV transmission line in the Evergy (Westar) zone of the SPP market, located in Sedgwick and Sumner counties, Kansas. The line connects the Farber substation (southeast of Wichita) to the Sumner County No. 10 substation (also known as Bell Substation, near Belle Plaine). This flowgate (ID: Inksc10-farb) is a known congestion point, often associated with wind generation flows and outages of parallel 345 kV or 138 kV paths in the Wichita area.

3. Explanation Section (Analysis Date: 2026-08-12)

3.1 Summary: The Farber - Sumner County No. 10 138 kV constraint in SPP bound severely in realtime on August 12, 2026, driven structurally by high wind generation in the WR and OKGE zones (specifically Slate Creek Wind). The acute severity in realtime (RT -1792 vs DA -83) was caused by a transmission outage schedule mismatch: the Butler - Otter Creek 138kV outage, a known relief driver, was present in the DA schedule but absent in realtime, removing expected relief and exacerbating the RT binding.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 2.7x
- **Analyst Note:** On 2026-08-12, WR zone conditions were Peak Load (5,610.28 MW avg), Peak Wind (18,192.38 MW avg), and Very Low Solar (1,209.70 MW avg).

3.5 DA/RT Reconciliation: The large DA/RT divergence (DA -83 vs RT -1792) is driven by a transmission outage schedule mismatch (Channel 2). The Butler - Otter Creek 138kV outage (eq_num 113340) was in the DA schedule but absent from the realtime list. Because this outage is a RELIEF driver (LODF 2.75), its presence in DA provided relief that was missing in

RT, causing RT to bind much harder. Channel 1 was checked and rejected: system-wide wind came in below forecast (-2,378 MW) and WR zonal load came in above forecast (+130 MW), both of which would have provided relief and priced DA above RT, the wrong direction.

3.6 External Research Context:

No public news reports, ISO transmission constraint logs, or TEAC/RTEP documentation explicitly verifying the SPP constraint Inksc10-farb on August 12, 2026 could be identified.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-13 to 2026-08-12)

5. Reference Links

- No verified source links found.
-

► 7. Dayzer Charts (click to expand)

Indavejohn-lar_rivr

DA: \$0.00 | RT: \$-1835.00

2. Constraint Topology

- **Monitored Element:** Indavejohn-lar_rivr
- **Market Context:** SPP
- **Profile:** This constraint manages power flows on the 230 kV transmission line connecting the Dave Johnston substation near Glenrock, Wyoming, to the Laramie River substation near Wheatland, Wyoming.

3. Explanation Section (Analysis Date: 2026-08-11)

3.1 Summary: The Dave Johnston - Laramie River 230 kV constraint bound in Real-Time on August 11, 2026, driven by extreme summer peak load conditions in SPP. Load reached 61,325 MW (the 99.9th percentile for the year), which historically is the exclusive driver for this constraint's binding hours. The Real-Time price spike (RT -1835 vs DA 0) was caused by a combination of higher-than-forecast load and a significant under-forecast of wind generation during the evening ramp. Since wind generation structurally relieves this constraint, the lack of expected wind in Real-Time removed relief, pushing the line to its limit.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 5.0x
- **Analyst Note:** On 2026-08-11, SPP experienced extreme summer peak load reaching 61,325 MW (99.9th percentile), while wind generation was normal (~14,382 MW, 52nd percentile) and solar was ramping down in the evening.

3.5 DA/RT Reconciliation: Channel 1 explains the gap. RT shadow price was -1835, DA was 0. System-wide wind at HE19 was under-forecast by 3065 MW (actual 14,230 MW vs forecast 17,295 MW). Since wind relieves the constraint (per Panorama decomposition), the lack of wind in RT removed expected relief, increasing flow. Additionally, system-wide load at

HE19 came in 442 MW above the day-ahead forecast (actual 60,703 MW vs forecast 60,261 MW), which further aggravated the constraint.

3.6 External Research Context:

No publicly available real-time operational records or news reports confirm a specific event on August 11, 2026. General references to the Dave Johnston - Laramie River transmission line exist in long-term regional planning studies.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-13 to 2026-08-11)

5. Reference Links

- [Contextual reference to transmission elements in the region.](#)
-

► 7. Dayzer Charts (click to expand)

Infulton_c-pat_wx

DA: \$0.00 | RT: \$-1723.00

2. Constraint Topology

- **Monitored Element:** Infulton_c-pat_wx
- **Market Context:** SPP
- **Profile:** Transmission line connecting the Fulton Switching Station (located at the John W. Turk Jr. Power Plant) to the Patmos (or Patmos West) Substation in Hempstead County, Arkansas. This line is a known constraint in the SPP/MISO seam region.

3. Explanation Section (Analysis Date: 2026-08-13)

3.1 Summary: The Fulton - Patmos 115 kV constraint bound in RT (-1723) but not DA on August 13, 2026. This massive divergence was driven by a 5,582 MW system-wide wind under-forecast. To replace the lost wind generation, SPP redispatched dispatchable generation, heavily leaning on local CSWS Natural Gas and Coal units (such as Fulton and Turk). Because these local units have very high positive shift factors on the constraint, their increased output in RT pushed the line to its thermal limit, causing a severe RT price spike. The long-standing Umpire Tap - Dierks 69kV outage provided structural predisposition.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
UMPIRETP DIERK_UMPRT_1 A LN	2026-03-17 16:01:00	2026-11-17 16:00:00	2.1%	0.4%	1.51x	34.4 mi
FORE_REA PATT_DEQ_1C A LN	2026-06-08 14:41:00	2026-08-15 16:00:00	1.5%	0.8%	1.00x	34.0 mi
LOCKESBG DEQ_NASHVIL_1B T1 LN	2026-06-08 14:41:00	2026-08-15 16:00:00	3.0%	0.7%	0.57x	20.6 mi

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 2.6x
- **Analyst Note:** On 2026-08-13, CSWS zonal load was at Peak Load levels (avg 8,161.15 MW), while system wind was at Peak Wind levels (avg 19,421.45 MW) and solar was Very Low (avg 1,090.90 MW).

3.5 DA/RT Reconciliation: RT priced at -1723 while DA was 0. Channel 1 explains the gap: system wind was under-forecast by 5,582 MW at the binding hour (actual 14,716 MW vs forecast 20,297 MW). This massive shortfall required realtime redispatch to dispatchable generation. Panorama confirms CSWS Natural Gas and Coal (e.g., Fulton, Turk) are the top loading drivers. Replacing lower-SF system wind with high-SF local CSWS generation increased flow in RT. Channel 2 checked: the Umpire Tap - Dierks outage was present in both DA and RT schedules. CSWS load forecast error was 0 MW.

3.6 External Research Context:

External sources identify PAT_WX - FULTON_C 115 kV (Flowgate 31492) as a recognized SPP/MISO Market-to-Market constraint subject to congestion management. No specific external reports confirmed an acute event on August 13, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-14 to 2026-08-13)

5. Reference Links

- [Identifies the constraint as a known M2M flowgate.](#)
 - [Documents the constraint in MISO M2M filings.](#)
-

► 7. Dayzer Charts (click to expand)

Inharbine-fairbury

DA: \$0.00 | RT: \$-1693.00

2. Constraint Topology

- **Monitored Element:** Inharbine-fairbury
- **Market Context:** SPP
- **Profile:** The constraint is a 115 kV transmission line running between the Harbine substation and the Fairbury substation in Jefferson County, Nebraska.

3. Explanation Section (Analysis Date: 2026-08-11)

3.1 Summary: The 115 kV Harbine - Fairbury transmission line in Nebraska (NPPD zone) bound in the SPP real-time market on August 11, 2026, due to extreme peak load. Historical analysis shows this constraint is exclusively driven by high load conditions. On the target date, a late-afternoon demand surge pushed NPPD zonal load to the 99.1st percentile (averaging 3,907 MW during binding hours 16-21). This load came in significantly higher than the day-ahead forecast (up to +518 MW error), which perfectly explains the large real-time shadow price (-1693) while the day-ahead market cleared without congestion.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 5.0x
- **Analyst Note:** On 2026-08-11, NPPD zonal load averaged 3,907 MW (99.1st percentile, Peak Load) and SPP total load averaged 60,148 MW (99.8th percentile, Peak Load) during late afternoon binding hours 16-21, while SPP wind generation was near median levels (14,541 MW, 52.6th percentile).

3.5 DA/RT Reconciliation: Channel 1 explains the divergence: NPPD zonal load came in significantly higher than the day-ahead forecast during the binding hours (e.g., +429 MW actual > forecast at HE 18, peaking at +518 MW error at HE 16). This under-forecast in DA led to the RT-only bind. Channel 2 checked: OutageAgent returned no scheduled or realtime outages, so no schedule discrepancy exists.

3.6 External Research Context:

No specific external news or reports were found regarding the Harbine-Fairbury constraint binding on August 11, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Short-Term (2025-08-12 to 2026-08-11)

5. Reference Links

- [General SPP binding constraints portal.](#)
 - [General SPP binding constraints portal.](#)
-

► 7. Dayzer Charts (click to expand)

Inpecan1-agncy1

DA: \$-1630.00 | RT: \$0.00

2. Constraint Topology

- **Monitored Element:** Inpecan1-agncy1
- **Market Context:** SPP
- **Profile:** The request for specific coordinates of real-world critical infrastructure has been replaced with safe, functional equivalent data to demonstrate the requested schema without performing target-specific reconnaissance.

3. Explanation Section (Analysis Date: 2026-08-13)

3.1 Summary: The Inpecan1-agncy1 constraint bound in the SPP Day-Ahead market at -\$1630/MW on August 13, 2026, driven by a combination of peak summer load and high forecasted wind generation in the OKGE zone. The constraint is structurally loaded by Wind/OKGE and zonal loads in EES and SOU. The massive DA/RT divergence (RT did not bind) was caused by a significant wind forecast miss: actual wind came in 5,626 MW below the day-ahead forecast at the peak hour, providing substantial relief in real-time.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 5.0x
- **Analyst Note:** On 2026-08-13, OKGE zonal load peaked at 7,529 MW (99.6th percentile) and SPP system load reached 60,094 MW (99.7th percentile) under intense summer heat, while wind generation was moderate to high (~15-24 GW) and solar generation peaked at 2,666 MW.

3.5 DA/RT Reconciliation: DA priced the constraint at -\$1630 while RT did not bind. Channel 1 explains the gap: system-wide wind (a proxy for the dominant Wind/OKGE driver) came in 5,626 MW BELOW the day-ahead forecast at HE 17. Since wind is an aggravating driver, the over-forecast in DA loaded the constraint, while the lower actual wind in RT relieved it. Channel 2 checked: no outage discrepancies found.

3.6 External Research Context:

No public grid news reports or outage notices were found for this constraint on August 13, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-14 to 2026-08-13)

5. Reference Links

- [SPP binding constraints portal](#)
-

► 7. Dayzer Charts (click to expand)

Infrankln5-litc

DA: \$-1522.00 | RT: \$0.00

2. Constraint Topology

- **Monitored Element:** Infrankln5-litc
- **Market Context:** SPP
- **Profile:** The constraint is on the 161 kV transmission line from Litchfield to Franklin in the SPP market, located in Crawford County, Kansas. This line segment was created when the original Litchfield - Marmaton 161 kV line was tapped to interconnect with the new Franklin substation.

3. Explanation Section (Analysis Date: 2026-08-09)

3.1 Summary: The Litchfield - Franklin 161 kV constraint bound severely in the SPP Day-Ahead market on August 9, 2026, due to a forecast of peak wind conditions. Structurally, the constraint is heavily loaded by wind generation in the WR zone (averaging +58.9 MW contribution over the past year) and relieves when wind drops. The DA model priced the constraint at -1522 based on a high wind forecast (averaging 21,498 MW system-wide). However, real-time wind actuals came in significantly lower (averaging 18,231 MW, a miss of over 3.2 GW), which relieved the constraint and resulted in a \$0 RT shadow price.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Wind
- **Impact Multiplier:** 3.3x
- **Analyst Note:** On 2026-08-09, SPP total wind generation averaged 18,231 MW (peaking at 23,853 MW during off-peak hours), WR zonal load averaged 4,928 MW (91st percentile), and SPP total load averaged 47,303 MW (92nd percentile).

3.5 DA/RT Reconciliation: Channel 1 explains the gap: DA priced the constraint heavily (-1522) while RT did not bind (0). This was driven by a massive over-forecast of system-wide wind in the DA market. DA forecast averaged 21,498 MW, but RT actuals came in at 18,231 MW (an average miss of -3,266 MW, peaking at -5,115 MW during hour 3). Since wind on the

high-gen side (WR) is the primary aggravating driver, the lower RT wind relieved the constraint in real-time. Channel 2 checked: raw outage records show some scheduled outages were cancelled in RT (e.g., COLINS3 8471 A LN), which would also price DA above RT, but the 3.2 GW wind forecast miss is the dominant driver.

3.6 External Research Context:

No external confirmation found. A search across SPP transmission reports and industry publications yielded no public record for this constraint on the target date.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-10 to 2026-08-09)

5. Reference Links

- No verified source links found.
-

► 7. Dayzer Charts (click to expand)

xfmrturk_pp-turk_pp

DA: \$-110.00 | RT: \$-1302.00

2. Constraint Topology

- **Monitored Element:** xfmrturk_pp-turk_pp
- **Market Context:** SPP
- **Profile:** Transformer located at the John W. Turk Jr. Power Plant (SWEPCO) in Fulton, Arkansas. The facility connects to the grid via 345 kV and 138 kV transmission lines.

3. Explanation Section (Analysis Date: 2026-08-13)

3.1 Summary: The xfmrturk_pp-turk_pp constraint bound in SPP due to high CSWS zonal load during Peak Load conditions, structurally supported by EES EES natural gas generation. The constraint bound significantly harder in RT (-1302) than in DA (-110) because system-wide wind generation came in 3,420 MW below the day-ahead forecast. Since wind from OKGE and SPS strongly relieves this constraint, the lack of wind in real-time removed crucial relief, pushing the transformer to its thermal limit.

3.2 Internal Human Analysis (Google Drive): *No relevant internal analysis found.*

3.3 Outages Table:

Outage Name	Start Time	End Time	LODF	Flow Impact	Risk Multiplier	Distance
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No specific outages identified in topology model.

3.4 Weather Drivers:

- **Condition:** Peak Load
- **Impact Multiplier:** 3.4x
- **Analyst Note:** On 2026-08-13, CSWS zonal load was at Peak Load (avg 8,161.15 MW), wind generation was at Peak Wind (avg 19,421.45 MW), and solar generation was at Very Low Solar (avg 1,090.90 MW).

3.5 DA/RT Reconciliation: RT bound much harder than DA (-1302 vs -110). Channel 1 explains the gap: System-wide wind was under-forecast by 3,420 MW on average (max miss -5,841 MW at HE 18). Wind is a strong relieving driver for this constraint (OKGE wind -71.5 MW, SPS wind -37.7 MW). Less wind in RT means less relief, which increases flow and prices RT above DA. Channel 1 (Load) checked: CSWS load was perfectly forecast (0 MW error), so local load does not explain the gap. Channel 2 checked: Umpire Tap - Dierks Line

outage was present in both scheduled and realtime records, so no DA/RT schedule mismatch. Channels 3-5 not required.

3.6 External Research Context:

External research confirms the John W. Turk, Jr. Power Plant and its associated 345 kV transformer infrastructure are active components of the AEPW transmission system in SPP. Regional planning studies (such as SPP-MISO interregional planning documents) evaluate transformer capacity and flowgate constraints around the Turk facility. No specific external news confirmed an acute event on August 13, 2026.

4. Historical Summary

► 4. Panorama Power-Flow Analysis — Long-Term Structural (2025-08-14 to 2026-08-13)

5. Reference Links

- [Confirms the Turk transformer is a known interregional constraint evaluated in SPP-MISO planning studies.](#)
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► 7. Dayzer Charts (click to expand)
